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Where I Becomes Infinite

Foundation Models Overview

Module 1.2: LLM Fundamentals | Prof. Sashikumaar Ganesan |
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Session Overview

What You Will Learn

- What a foundation model is
- How foundation models differ from earlier AI
- Key capabilities and limitations
- Representative examples in 2024–2025

Concept at a Glance

- **Concept ID:** AI-LM-FN-TH-000005
- **Duration:** 8 minutes
- **Bloom's Level:** B2 – Understand
- **Difficulty:** L1 – Beginner
- **Module:** 1.2 LLM Fundamentals



Learning Objectives

By the End of This Session You Will Be Able To

1. **Define** a foundation model and describe its core characteristics
2. **Explain** how large-scale pre-training creates general-purpose representations
3. **Identify** major foundation models by name, organisation, and modality
4. **Distinguish** foundation models from task-specific and classical ML models

Prerequisites & Context



Prerequisites

- What is a Language Model? (AI-LM-FN-TH-000001)
- Probability and Token Prediction (AI-LM-FN-TH-000002)
- Autoregressive Generation (AI-LM-FN-TH-000003)
- Model Sizing and Scaling Laws (AI-LM-FN-TH-000004)

What This Unlocks

- Comparing GPT, Llama, TinyLlama (AI-LM-FN-AI-000001)
- Transformer architecture deep-dive (Module 2)
- Fine-tuning and RLHF workflows (Module 7)
- Agentic system design (Module 5)



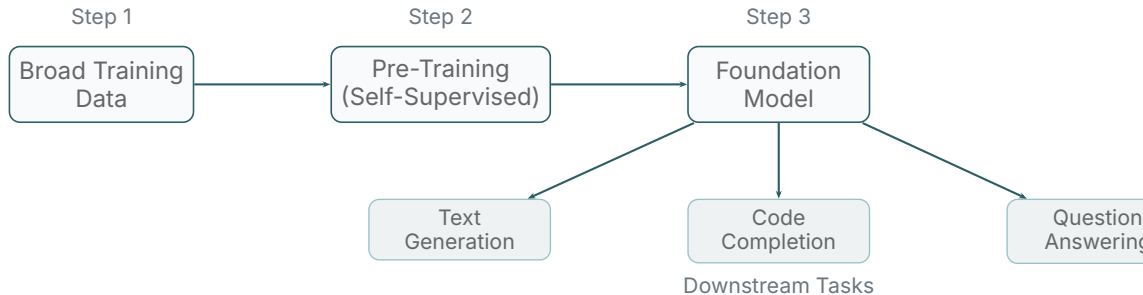
What is a Foundation Model?

Definition

A **foundation model** is a large AI model trained on broad data at scale that can be *adapted* to a wide range of downstream tasks through fine-tuning, prompting, or integration with external tools.

- Coined by Stanford CRFM in 2021
- **Scale:** billions to trillions of parameters
- **Breadth:** trained on diverse, internet-scale corpora
- **Adaptability:** one model, many tasks (translation, coding, reasoning ...)
- **Emergent abilities:** capabilities that arise only at large scale

Foundation Model Paradigm --- Visual Overview





From Task-Specific to Foundation Models

Property	Classical ML / Task-Specific	Foundation Model
Training data	Narrow, labelled	Broad, largely unlabelled
Parameters	Thousands – millions	Billions – trillions
Tasks supported	One	Hundreds
Adaptation method	Retrain from scratch	Fine-tune or prompt
Emergent abilities	Rare / none	Demonstrated at scale
Examples	ResNet, SVM	GPT-4, Llama 3, Claude



Core Characteristics of Foundation Models --- Part 1

Scale, Self-Supervision, and Emergence

- **Massive scale:** GPT-4 estimated at ~ 1 trillion parameters; training costs millions of USD
- **Self-supervised pre-training:** no manual labels needed — models predict masked or next tokens
- **Transfer learning:** pre-trained weights carry general knowledge reusable across tasks
- **Emergent abilities:** chain-of-thought reasoning, few-shot learning appear only at large scale
- **In-context learning:** tasks solved by examples in the prompt without weight updates



Core Characteristics of Foundation Models --- Part 2

Multimodality, Alignment, and Limitations

- **Multimodal potential:** modern foundation models handle text, images, audio, code
- **RLHF alignment:** Reinforcement Learning from Human Feedback steers behaviour safely
- **Hallucination:** models can produce confident but factually wrong outputs
- **Knowledge cutoff:** static training data — world knowledge stops at a fixed date
- **Compute cost:** inference and training require significant GPU/TPU infrastructure



Major Foundation Models --- Text & Language

Model	Organisation	Released	Notable Feature
GPT-4o	OpenAI	2024	Multimodal (text + image + audio)
Claude 3.5	Anthropic	2024	Long context (200k tokens)
Gemini 1.5	Google Deep-Mind	2024	1M+ token context window
Llama 3	Meta	2024	Open weights, 8B–70B sizes
Mistral 7B	Mistral AI	2023	Efficient open model
TinyLlama	Community	2024	1.1B – edge deployment



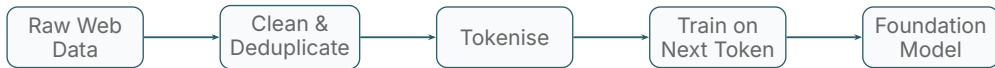
Foundation Models Beyond Text

Modality Expansion

- **Vision:** CLIP (OpenAI) aligns images and text; DALL-E 3 generates images from prompts
- **Audio:** Whisper (OpenAI) transcribes speech; AudioLM generates realistic audio
- **Code:** Codex, StarCoder, DeepSeek-Coder specialise in programming tasks
- **Science:** AlphaFold 2/3 for protein structure; Genie for climate modelling
- **Video:** Sora (OpenAI) generates video from text descriptions



Pre-Training Pipeline



Pipeline Stages

- **Raw data:** Common Crawl, books, code, scientific papers — trillions of tokens
- **Clean & deduplicate:** remove duplicates, toxic content, low-quality pages
- **Tokenise:** convert text to integer IDs (BPE, SentencePiece)
- **Train:** minimise cross-entropy loss over billions of steps on GPUs/TPUs



Emergent Abilities --- What Scale Unlocks

Abilities Not Present in Smaller Models

- **Few-shot learning:** solve new tasks from 3–5 examples in the prompt
- **Chain-of-thought reasoning:** step-by-step arithmetic and logic
- **Multi-step planning:** decompose complex goals autonomously
- **Code generation:** write, debug, and explain programs from natural language
- **Cross-lingual transfer:** translate between hundreds of languages without explicit training

GenAI Application --- Chatbots & Virtual Assistants



Production Use Case

How it works: A foundation model (e.g. GPT-4) is fine-tuned with RLHF to follow instructions and be helpful and safe. It is then served via an API with a system prompt that customises its persona.

- **Examples:** ChatGPT, Claude, Gemini Advanced, Copilot
- **Scale:** hundreds of millions of daily active users
- **Tasks:** drafting emails, answering questions, coding assistance, tutoring
- **Infrastructure:** NVIDIA H100/A100 clusters, vLLM/TGI serving frameworks

GenAI Application --- Code Intelligence



Foundation Models for Software Development

How it works: Models pre-trained on GitHub code (billions of tokens) learn syntax, semantics, and common patterns across dozens of programming languages.

- **Examples:** GitHub Copilot (GPT-4), Cursor, Amazon CodeWhisperer
- **Tasks:** autocomplete, test generation, refactoring, documentation
- **Adoption:** over 1 million developers using Copilot (2024)
- **Productivity gain:** studies show 30–55% faster task completion

GenAI Application --- Enterprise Search & RAG



Retrieval-Augmented Generation

How it works: A foundation model is paired with a vector database. At query time, relevant documents are retrieved and injected into the prompt, grounding the response in real data.

- **Solves:** knowledge cutoff and hallucination limitations
- **Stacks:** LangChain + Pinecone + GPT-4 or Llama 3
- **Use cases:** legal document search, customer support, medical Q&A
- **Outcome:** accurate, citeable answers over proprietary knowledge bases

GenAI Application --- Scientific Discovery



Foundation Models in Research

How it works: Domain-specific foundation models trained on scientific corpora (papers, sequences, simulations) accelerate hypothesis generation and analysis.

- **AlphaFold 3:** protein–ligand structure prediction in seconds vs. months of wet lab work
- **ESM-2 (Meta):** protein language model with 15 billion parameters
- **BioMedLM:** clinical notes, drug interactions, literature summarisation
- **Impact:** Nature named AlphaFold “Method of the Year 2021”



GenAI Application --- Multimodal Agents

Foundation Models as Agent Brains

How it works: A vision-language foundation model receives screen/camera input, reasons about the current state, and outputs tool-use actions (click, type, search).

- **Examples:** GPT-4V agents, Claude Computer Use, Gemini 1.5 Pro
- **Tasks:** web browsing, data extraction, form filling, UI automation
- **Frameworks:** LangGraph, AutoGen, OpenAI Swarm
- **Enterprise value:** automate repetitive knowledge-worker tasks at scale



Open vs. Closed Foundation Models

Dimension	Open Weights	Closed / Proprietary
Access	Download and run locally	API only
Cost	Compute only	Per-token pricing
Privacy	Data stays on-premise	Data sent to provider
Customisation	Full fine-tuning freedom	Limited (some providers)
Performance	Increasingly competitive	Generally frontier
Examples	Llama 3, Mistral, Falcon	GPT-4, Claude, Gemini



Limitations & Risks of Foundation Models

Critical Awareness

- **Hallucination:** confident generation of false facts; requires grounding or verification
- **Bias:** training data biases propagate into model outputs and decisions
- **Privacy leakage:** models may memorise and reproduce training data
- **Environmental cost:** GPT-3 training emitted ~552 tonnes CO₂ equivalent
- **Safety alignment:** without RLHF, models can produce harmful or deceptive content

Skills You Are Building



Conceptual Skills

- Defining and classifying foundation models
- Understanding scale–capability relationships
- Recognising emergent ability patterns
- Evaluating open vs. closed model trade-offs

Applied Skills

- Selecting the right foundation model for a task
- Understanding API-based model access
- Identifying appropriate adaptation strategies
- Assessing risk vs. capability for deployment

Knowledge Graph Positioning



This Concept Requires

- What is a Language Model?
- Probability & Token Prediction
- Autoregressive Generation
- Model Sizing & Scaling Laws

This Concept Enables

- Foundation Model Comparison Lab
- Transformer Architecture (M2)
- RAG & Vector Databases (M5)
- Fine-tuning Workflows (M7)
- LLMOps & Deployment (M9)



Common Misconceptions

Clarifications

- **Misconception:** "Foundation models understand language like humans do."
Reality: They perform sophisticated pattern matching over statistical distributions.
- **Misconception:** "Bigger is always better."
Reality: Chinchilla scaling shows data quantity matters as much as parameter count.
- **Misconception:** "Open models are always safe to deploy."
Reality: Open weights still require alignment fine-tuning and safety evaluation.

Summary



Foundation Models Overview — Key Takeaways

1. A **foundation model** is a large model trained at scale on broad data and adapted to many tasks
2. **Self-supervised pre-training** on trillions of tokens gives rise to general-purpose representations
3. **Emergent abilities** (few-shot learning, reasoning) appear only at large scale
4. Major models include **GPT-4o, Claude 3.5, Gemini 1.5, Llama 3, Mistral**
5. **Limitations** include hallucination, bias, knowledge cutoff, and compute cost

Next Steps



Immediately After This Session

- Review the Foundation Model Comparison AI App (AI-LM-FN-AI-000001)
- Explore Hugging Face Model Hub — browse by parameter count
- Run a TinyLlama inference locally using Ollama

Coming Up in Module 2

- Self-Attention Mechanism (AI-TR-FN-TH-000002)
- Multi-Head Attention (AI-TR-FN-TH-000003)
- Positional Encoding (AI-TR-FN-TH-000004)
- Embeddings (AI-TR-FN-TH-000005)

Thank You

Foundation Models Overview | AI-LM-FN-TH-000005